

Ranadeep Biswas, Ph.D.

CS Researcher & Engineer
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EXPERIENCE

Informal Systems *July 2021 – Present*

Emerald Developer Sept 2025 – Present

Modular framework for institutional blockchains
Tendermint (Malachite), Simplex (Commonware) consensus engines.
Ethereum execution layer and Solidity contracts.
Projects: 1Money Network, Initiativ.

ibc-starknet Developer July 2024 – Aug 2025

IBC protocol implementation in Cairo for Starknet
Starknet stack, Cairo, zero-knowledge systems, production deployment.

ibc-rs Developer July 2023 – June 2024

Core IBC implementation in Rust for Cosmos ecosystem
Trustless interoperability stack, protocol engineering, open-source project management.

Blockchain Auditor Jan 2023 – June 2023

Security audits for Cosmos SDK, Tendermint, CosmWasm protocols
DeFi attack vectors, protocol design patterns, formal audit methodology.
Projects: Gravity Bridge, IBC Spec, Osmosis, Mars Protocol, Namada, Axelar.

Testing Engineer July 2021 – Dec 2022

Specifications and formal verification for blockchain protocols
TLA+, Apalache model checker, model-based testing, test suite design.

EDUCATION

Ph.D. in Computer Science Nov 2017 – March 2021

Université Paris Cité (IRIF), France
Thesis: *Automated Formal Testing of Storage Systems and Applications*

Master of Science in Computer Science Aug 2015 – April 2017

Chennai Mathematical Institute, India
Software verification using CBMC

Bachelor of Science in Mathematics and Computer Science Aug 2012 – April 2015

Chennai Mathematical Institute, India

SKILLS

Distributed Systems · Blockchain Interoperability · Byzantine Consensus · Automated Testing ·
Formal Methods · Smart Contracts · Open-source Contribution
Solidity · EVM · Starknet · Cairo · Cosmos SDK · IBC · Rust · TLA+ · Lean4 · Linux · Cloud
Infrastructure · AI Agents